School of Computer Science and Engineering

Faculty of Engineering & Technology

Jain Global Campus, Kanakapura Taluk – 562112 Ramanagara District, Karnataka, India

A Project Report on

"IARS Nanoswimmers Simulator"

Submitted by

Alok Singh – 22BTRAD003

Deeraj Reddy – 22BTRAD013

Sathvika Reddy - 22BTRAD024

Under the guidance of

Dr. S. Saravanakumar

CANDIDATE'S DECLARATION

We, Alok Singh(22BTRAD003), Deeraj Reddy(22BTRAD013) and Sathvika Reddy(22BTRAD024) students of VIIIth semester B.Tech in Computer Science and Engineering – Artificial Intelligence & Data Engineering, at School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed to-be University), hereby declare that the internship work titled **"IARS Nanoswimmers Simulator"** has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science and Engineering- Artificial Intelligence & Data Engineering** during the month of **April 2026**. Further, the matter presented in the work has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

School of Computer Science and Engineering
Faculty of Engineering & Technology
Jain Global Campus, Kanakapura Taluk - 562112
Ramanagara District, Karnataka, India

CERTIFICATE

This is to certify that the project work titled **"IARS Nanoswimmers Simulator"** is carried out by Alok Singh(22BTRAD003), Deeraj Reddy(22BTRAD013) and Sathvika Reddy(22BTRAD024)**,** students of Bachelor of Technology at the School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfilment for the award of degree in Bachelor of Technology in Computer Science and Engineering - Artificial Intelligence Data Engineering, during the year **2026**.

Dr. S. Saravanakumar

Project Coordinator

ABSTRACT

This project presents the design and implementation of a two dimensional computational simulation framework that models the initial decision making stages of an Individual Adaptive Regulation Strategy inspired nanoswimmer system. The central aim of this work is to understand how artificial nanoswimmers can independently sense biochemical information from their environment, assess the quality of that information, and then collectively converge on a reliable navigation direction toward a tumour like target region under controlled conditions.

The simulated environment is constructed as a simplified Biological Gradient Field that mimics tumour induced biochemical signalling. This field is modelled with the highest signal intensity concentrated at the centre of the domain, representing the tumour core, and a smooth radial decay in intensity as the distance from the centre increases. Such a formulation captures the essential spatial characteristics of real biological gradients while remaining computationally tractable for early stage algorithm development and visualization. Within this gradient field, a population of nanoswimmers is randomly initialized across the domain. Each nanoswimmer operates as an independent sensing agent with no prior global knowledge of the environment. To estimate local directional cues, each agent samples the gradient field using a finite difference based sensing approach. By probing small spatial offsets around its current position, a nanoswimmer approximates the local gradient direction, which serves as its individual estimate of where the tumour signal is increasing. This process intentionally introduces sensing noise and variability, reflecting the uncertainty that would be present at the nanoscale in real biological environments.

Rather than relying on all agents equally, the system identifies a subset of elite nanoswimmers that exhibit the highest fitness values. These elite agents are assumed to possess superior sensing quality at that simulation step. Their individually estimated movement directions are then combined using a weighted vector based averaging method, where each direction contributes proportionally according to its associated fitness. This fusion process filters out noisy or misleading estimates from poorly positioned agents and yields a single stable global navigation direction. Importantly, this direction represents collective swarm intelligence rather than the decision of any single nanoswimmer.

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IARS Nanoswimmers Simulator

1. Introduction

1.1 Background

Nanoswimmers are tiny artificial agents that operate at micro and nanoscale levels inside fluid environments, where the rules of physics are very different from what we experience at larger scales. At this level, the Reynolds number is extremely low, which means viscous forces dominate while inertial effects are almost negligible. Because of this, movement becomes highly restricted and sensitive to disturbances, with even small random effects like Brownian motion having a noticeable impact on how these agents behave. Unlike traditional robotic systems, nanoswimmers cannot depend on centralized control, GPS-like positioning, or precise actuation. They function under strict limitations such as limited sensing ability, noisy environmental signals, and minimal computational capacity. These constraints make it necessary to rely on decentralized, adaptive approaches that are often inspired by biological systems rather than conventional engineering methods.

In the context of biomedical applications, nanoswimmers offer promising possibilities in areas like targeted drug delivery, tumour localization, minimally invasive diagnostics, and precision medicine. Their ability to move through complex biological environments such as blood vessels and tissues makes them especially useful for delivering treatment directly to specific regions. Biological systems naturally contain spatial gradients, including variations in chemical concentration, oxygen levels, temperature, and pH. Tumours create particularly strong gradient fields due to irregular metabolism, low oxygen conditions, and abnormal blood vessel formation. These gradients carry directional information that nanoswimmers can use to estimate where the tumour is located. Building on this idea, the project moves beyond simple gradient sensing by also incorporating motion, fluid flow, and target-seeking behaviour, making the simulation more aligned with realistic biological conditions.

1.2 Evolution of Adaptive Navigation Strategies

Early nanoswimmers navigation approaches relied on passive mechanisms or external control systems such as magnetic fields. While these methods provide directional control, they suffer from major limitations including lack of adaptability, dependence on prior knowledge, and inability to scale efficiently.

To overcome these issues, swarm intelligence-based approaches were introduced. These methods treat nanoswimmers as a collective system where global intelligence emerges from local interactions.

The Individual Adaptive Regulation Strategy (IARS) builds upon this concept by introducing:

- Local sensing
- Reliability-based filtering
- Collective decision-making

In this project, IARS is **extended further** by integrating:

- Dynamic motion modelling
- Target-driven navigation
- Fluid flow influence

This creates a more **realistic hybrid model** combining:

- Gradient-based intelligence
- Physical motion dynamics

1.3 Motivation for the Project

While theoretical frameworks like IARS offer a strong conceptual understanding, they need to be tested in controlled simulation environments to evaluate how they actually perform. Earlier work mainly focused on static sensing and decision making, where agents estimate direction without physically moving. However, real nanoswimmers operate in far more complex conditions, including:

- Dynamic environments
- Flow driven systems
- Multi target scenarios

Because of this gap, the motivation behind this project is twofold:

- Validate the IARS decision making pipeline
- Extend it to a more realistic motion based simulation

By bringing together sensing, selection, and motion into a single framework, the project connects theoretical models with practical behaviour. It effectively bridges the gap between abstract algorithmic concepts and how nanoswimmers would function in real nanoscale environments.

2. Objective of Project

The objective of this project is to design and validate a complete nanoswimmers simulation framework that brings together multiple interconnected components, each playing a role in how agents sense their surroundings, evaluate information, and eventually decide where to move. The focus is not just on movement, but on understanding how a reliable decision emerges step by step from simple local observations to a coordinated group response.

2.1 Gradient Based Environment

The first step is to create a gradient based environment that represents tumour induced biological signals. This is done by modelling a field where the signal is strongest at the centre and gradually weakens as the distance increases. This setup reflects how real biochemical signals spread in the body. The purpose of this environment is to give nanoswimmers a direction to follow, based purely on signal variation, without needing any global map or external guidance.

2.2 Swarm Initialization

Nanoswimmers are initially placed randomly across the environment to simulate real deployment conditions. In practice, these agents would not start from perfect positions, so randomness helps recreate that uncertainty. Each nanoswimmer begins with no prior knowledge and acts independently. This setup allows us to observe whether the system can still produce a meaningful outcome even when starting from scattered and unpredictable positions.

2.3 Gradient Estimation

Each nanoswimmer estimates the direction of increasing signal by sampling nearby points in its surroundings. Instead of having access to exact information, it compares values at small offsets to understand how the signal changes locally. This method introduces slight inaccuracies, which is intentional

because real nanoscale sensing is never perfect. Through this process, every agent forms its own idea of the correct direction based only on what it can observe nearby.

2.4 Fitness Based Evaluation

Once a direction is estimated, each agent is assigned a fitness value based on how strong the signal is at its current position. This value acts as a measure of confidence. Agents located closer to the centre experience stronger signals and are more likely to make accurate estimates. By assigning a numerical score to each agent, the system can identify which ones are more reliable and which ones are likely influenced by noise.

2.5 Elite Selection Mechanism

Rather than treating all agents equally, the system selects only the top performing ones based on their fitness values. These selected agents are considered more trustworthy because they are operating in regions with clearer signals. By filtering out weaker agents, the system avoids being influenced by noisy or misleading information. This step improves the overall quality of the decision by focusing only on the most reliable inputs.

2.6 Collective Decision Formation

The selected agents then contribute their estimated directions, which are combined into a single global direction. Each direction is represented as a vector and weighted according to its fitness, meaning stronger and more reliable agents have a greater influence. When all these contributions are merged, the result is a stable direction that reflects the collective intelligence of the group rather than any single agent's decision.

2.7 Dynamic Motion Simulation

The framework also includes realistic motion behaviour to make the simulation closer to real conditions. Nanoswimmers move by combining multiple influences, including attraction toward targets, the effect of fluid flow, and a helical motion pattern. Target attraction pulls agents toward regions of interest, fluid flow introduces environmental forces that slightly alter their paths, and helical motion adds a natural oscillation to their movement. Together, these factors create more realistic trajectories.

2.8 Multi Target Navigation

Finally, the system supports navigation toward multiple targets instead of just one. Each nanoswimmer identifies the closest target and adjusts its direction accordingly. This makes the system more practical, as real biological environments often contain multiple regions of interest. It also shows that the system can handle more complex scenarios while still maintaining stable and coordinated behaviour.

3. Tools and Technologies Used

This project relies on a combination of programming tools and computational techniques to build, test, and visualize the nanoswimmers system. Each tool plays a specific role in handling different parts of the simulation, from numerical computation to visual interpretation.

3.1 Python

Python is used as the main programming language for the entire simulation. It provides a simple and flexible way to implement algorithms, handle mathematical operations, and manage the overall workflow. Its readability also makes it easier to experiment with different ideas and refine the model step by step.

3.2 NumPy

NumPy is used for handling numerical computations efficiently. Since the simulation involves many agents operating at the same time, vectorized operations help process data in parallel instead of using slow loops. It is mainly used for:

- Vector operations, where directions and movements are represented mathematically
- Distance computation, to measure how far agents are from targets
- Gradient estimation, where nearby values are compared to detect direction changes

3.3 Matplotlib

Matplotlib is used to visualize the simulation and make the results easier to understand. Instead of just working with numbers, it helps display how agents move and how the environment behaves. It is used for:

- Visualization of the gradient field and agent positions
- Animation to show how nanoswimmers move over time
- Vector field representation, where directions are shown using arrows

3.4 Jupyter Notebook

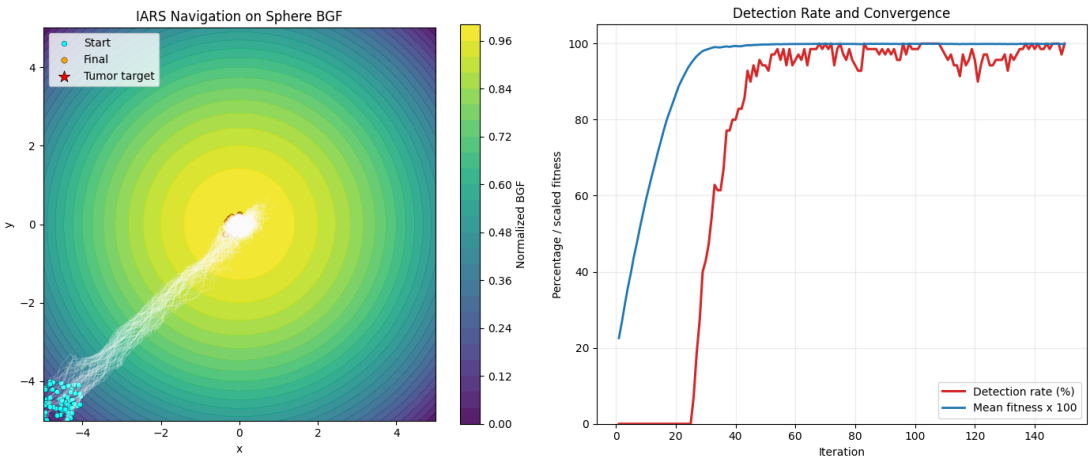
Jupyter Notebook is used as the development environment for building and testing the simulation. It allows the work to be done in small steps, making it easier to debug and validate each part of the system. It supports:

- Experimentation with different parameters
- Step wise validation of each stage in the pipeline

3.5 Algorithms Implemented

The core of the system lies in the algorithms that allow nanoswimmers to sense, evaluate, and decide collectively.

IARS Framework



The Individual Adaptive Regulation Strategy forms the base of the system. It focuses on how multiple agents independently sense information and then combine their inputs to form a reliable collective decision.

Finite Difference Gradient Estimation

$\partial U/\partial x \approx (U(x + \delta, y) - U(x - \delta, y)) / 2\delta$

This formula is used to estimate how the signal changes in the x direction. Here, U(x,y) represents the signal strength at position (x,y), and δ is a small step size used to sample nearby points. Since the exact gradient is not known, the system approximates it by checking how the value changes slightly to the left and right of the current position. This gives an estimate of the slope, which tells the agent which direction has increasing signal.

Fitness Function

$f_i = U(x_i, y_i)$

The fitness function assigns a value to each nanoswimmer based on its position. Here, (xi, yi) represents the location of the ith agent, and U(xi, yi) gives the signal strength at that point. A higher fitness value means

the agent is closer to the target and is likely making a more reliable decision. This value acts as a measure of confidence.

Elite Selection

$E = \rho N$

This formula defines how many agents are selected as elite. N is the total number of nanoswimmers, and ρ is a fraction between 0 and 1. By multiplying them, we get the number of top agents to keep. This step ensures that only the most reliable agents contribute to the final decision, reducing the effect of noise.

Weighted Vector Fusion

$V = \Sigma(i \in E)\ f_i\ v_i$

In this step, each selected agent contributes a direction vector vi, which represents where it believes the signal is increasing. The fitness value fi acts as a weight, meaning more reliable agents have a stronger influence. All these weighted vectors are summed to produce a single vector V, which represents the collective direction of the swarm. This is how multiple uncertain inputs are combined into a stable decision.

Nearest Target Selection (NEW)

$\text{target}_i = \arg\min \|T - X_i\|$

This formula helps each agent choose the closest target. Here, Xi is the position of the agent, and T represents possible target locations. The expression $\|T - X_i\|$ calculates the distance between the agent and each target. The arg min operation selects the target with the smallest distance. This allows agents to focus on the nearest goal instead of all targets.

Velocity Model (NEW)

$V = 0.8 \cdot D + 0.5 \cdot F + H$

This equation defines how each nanoswimmer moves. The final velocity V is a combination of three components:

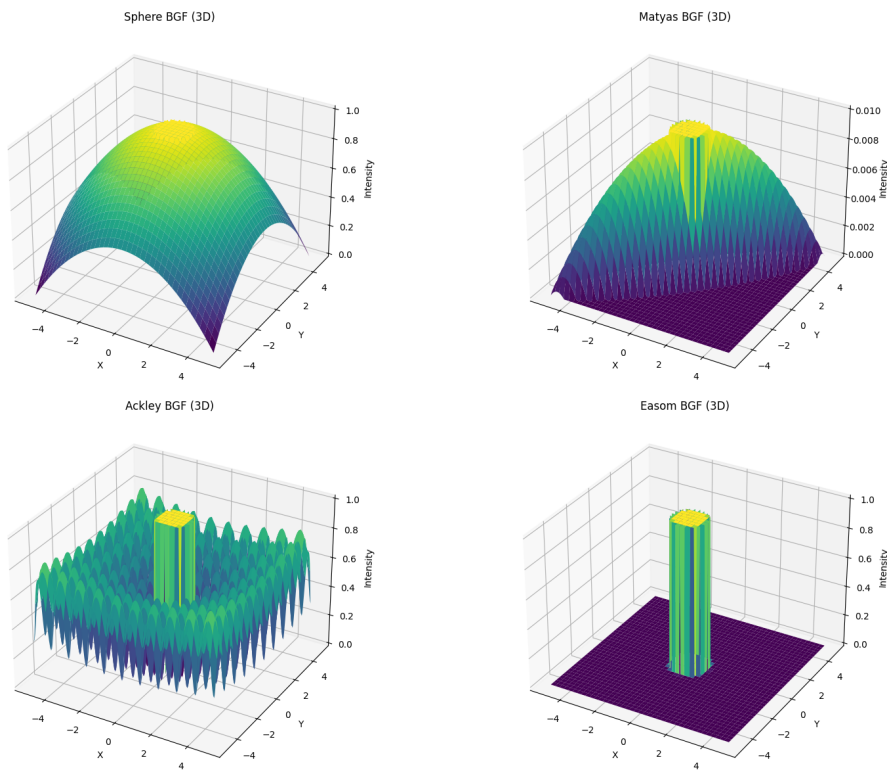
- D is the direction toward the target, which drives goal oriented movement
- F represents the blood flow vector, which models environmental influence
- H is the helical motion component, adding a natural oscillation to movement

The coefficients 0.8 and 0.5 control how strongly each factor influences motion. This combination ensures that movement is not purely straight or predictable, but instead reflects realistic behaviour where agents are influenced by both internal decisions and external forces.

4. System Architecture

The system architecture represents the logical and computational structure of the nanoswimmer simulation. It is designed as a layered pipeline in which information flows sequentially from environmental modelling to collective decision formation. Each layer is modular and clearly defined, allowing the sensing and decision making logic of the IARS framework to be validated independently before introducing motion or behavioural complexity.

4.1 Environment Layer



The Environment Layer defines the two dimensional simulation domain and models the tumour induced Biological Gradient Field. This field assigns a scalar signal intensity value to every point in the domain, with the maximum value located at the centre of the space to represent the tumour region. Signal intensity decreases smoothly as distance from the centre increases, forming a continuous and radially decaying gradient. This layer acts as a surrogate biological environment. It provides the ground truth signal

distribution that nanoswimmers attempt to sense and interpret. All gradient queries performed by nanoswimmers are routed through this layer, ensuring consistency and repeatability across simulations.

4.2 Nanoswimmers Agent Layer

The Nanoswimmer Agent Layer represents each nanoswimmer as an independent computational agent. Each agent maintains its own internal state, including spatial position, locally sensed gradient value, computed fitness score, estimated gradient direction, and corresponding direction vector. Agents do not directly communicate with one another. Instead, they operate under the assumption of limited local awareness, reflecting realistic nanoscale constraints. Their influence on the collective decision is mediated indirectly through the elite selection and fusion mechanisms, which operate at the swarm level rather than through explicit inter agent messaging.

4.3 Gradient Estimation Module

The Gradient Estimation Module enables each nanoswimmer to infer directional information from the biological gradient field. Using finite difference sampling, the agent probes the environment at small offsets around its current position. Differences in signal intensity between these sampled points are used to approximate the local gradient vector.

This module captures the essence of how nanoscale agents might sense environmental changes without direct access to global information. It intentionally allows estimation variability, which is later addressed through fitness evaluation and selective fusion.

4.4 Elite Selection Module

The Elite Selection Module evaluates all nanoswimmers based on their computed fitness values. Fitness reflects the strength of the biological signal detected at each agent's position and serves as a proxy for sensing reliability. Agents are ranked according to fitness, and only a subset with the highest values is selected as elite nanoswimmers. These elite agents are assumed to possess the most reliable gradient information and are therefore permitted to contribute to the collective navigation decision. This module is central to the IARS philosophy of prioritising quality over quantity in swarm level decision making.

4.5 Direction Fusion Module

The Direction Fusion Module aggregates directional estimates from the elite nanoswimmers to compute a single global navigation direction. Each elite agent's estimated direction is converted into a vector and weighted according to its fitness value. These weighted vectors are then combined through vector averaging to produce a stable and robust global direction. This fused output represents collective swarm intelligence and effectively suppresses noise originating from uncertain or poorly positioned agents.

4.6 Visualization Module

The Visualisation Module provides graphical representations of the simulation state and outputs. It displays the biological gradient field as a heatmap, nanoswimmer positions as scatter points, individual gradient directions as arrows, and the final fused global direction as a dominant vector. These visual outputs play a critical role in validating the correctness of the system architecture. They allow intuitive inspection of whether nanoswimmers are sensing meaningful gradients and whether the collective decision consistently points toward the tumour like region.

5. Implementation Logic and Code Explanation

This section explains the logical flow of the implemented simulation and describes how each code component contributes to the early computational stages of the Individual Adaptive Regulation Strategy. The implementation is intentionally kept minimal and interpretable so that the correctness of sensing, evaluation, and collective decision making can be verified without interference from motion dynamics or behavioural complexity.

5.1 Biological Gradient Field Definition

```
def bgf(x, y, alpha=0.01):
return 1 - alpha * (x**2 + y**2)
```

This function defines the Biological Gradient Field used throughout the simulation. The field has its maximum intensity at the centre of the two dimensional domain and decreases smoothly as the distance from the centre increases. This behaviour models tumour induced biological signals, where abnormal cellular activity produces strong local gradients that weaken with spatial separation. The quadratic distance term ensures radial symmetry, while the scaling parameter controls the rate of decay. The simplicity of this formulation is intentional. It allows the gradient structure to be analytically predictable, making it easier to validate whether nanoswimmers are sensing and responding correctly. All environmental sensing performed by agents relies on this function, making it the foundation of the simulation.

5.2 Nanoswimmer Initialization

```
positions = np.random.uniform(-10, 10, (N, 2))
```

Nanoswimmers are randomly initialized across the simulation domain to ensure unbiased and diverse environmental sampling. Each row in the array represents the two dimensional position of a nanoswimmer. Random initialization reflects realistic biological deployment scenarios, where nanoswimmers cannot be placed precisely and must operate under positional uncertainty. From an algorithmic perspective, this randomness is important because it tests whether the IARS decision logic remains stable regardless of initial conditions.

5.3 Gradient Approximation Using Finite Differences

```
gx = bgf(x + delta, y) - bgf(x - delta, y)
gy = bgf(x, y + delta) - bgf(x, y - delta)
theta = np.arctan2(gy, gx)
```

Each nanoswimmer estimates the local gradient direction using finite difference sampling. The agent evaluates the gradient field at small offsets around its current position along both axes. Differences in signal strength are used to approximate the partial derivatives of the field. This approach mimics how nanoscale agents might infer directional cues by comparing local concentration changes rather than accessing a global gradient map. The resulting angle represents the direction of steepest ascent in signal intensity. Importantly, this method introduces natural variability, which later stages of the IARS framework are designed to handle.

5.4 Fitness Computation

Fitness values are computed using the gradient field intensity at each nanoswimmer's position. A higher fitness indicates that the agent is located in a region with stronger biological signals and therefore has a more reliable gradient estimate. This step translates raw environmental sensing into a quantitative confidence measure. Fitness becomes the key regulator that determines which agents are trusted during collective decision making.

5.5 Elite Selection

```
elite_indices = np.argsort(fitness)[-elite_count:]
```

Elite selection is performed by ranking nanoswimmers according to their fitness values and selecting only the top performing subset. These elite agents are assumed to have the most reliable environmental information. This mechanism embodies the core idea of IARS. Instead of averaging information from all agents, which would amplify noise, the strategy prioritises quality over quantity. Poorly positioned nanoswimmers are excluded from influencing the global decision.

5.6 Direction Vector Construction

Directional angles from elite nanoswimmers are converted into unit vectors. This representation avoids discontinuities associated with angular values and allows directions to be combined geometrically. By working in vector space rather than angle space, the system ensures smooth aggregation even when individual directions differ slightly.

5.7 Weighted Direction Fusion

```
final_theta = np.arctan2(vy_f, vx_f)
```

In the fusion stage, elite direction vectors are weighted by their corresponding fitness values and summed. The resulting vector represents the collective navigation direction of the swarm. This weighted vector based fusion suppresses noisy estimates and amplifies strong, reliable signals. The final angle is computed from the fused vector components, yielding a stable global direction that points toward the tumour like region.

This step is critical because it transforms independent local sensing into collective swarm intelligence, which is the central objective of the IARS framework.

5.8 Helical Motion

$$H = (\cos\theta, \sin\theta)$$

This represents the natural oscillatory movement of nanoswimmers. The angle θ changes over time, and using cosine and sine converts this angle into a circular motion pattern. This adds a slight spiral or helical behaviour to the movement, making it more realistic instead of perfectly straight paths.

5.9 Visual Validation

Throughout the implementation, visual outputs are generated to validate correctness. Heatmaps confirm the structure of the gradient field, scatter plots show nanoswimmer distribution, and arrows illustrate individual and fused directions. These visualisations provide immediate qualitative confirmation that the sensing, evaluation, and fusion logic behaves as expected and consistently aligns the swarm toward the signal source.

6. Output

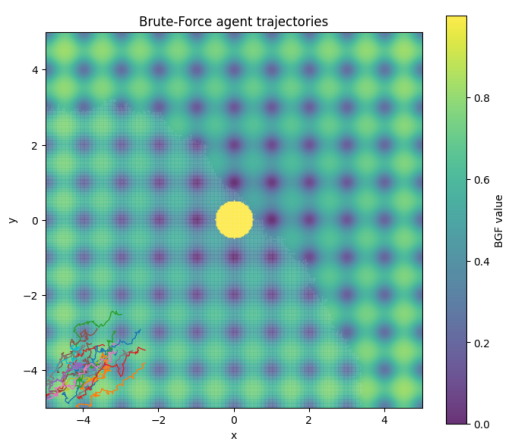
The simulation produces multiple visual and numerical outputs that together validate the correctness of the implemented sensing and decision making pipeline. Each output corresponds directly to a specific stage of the IARS inspired architecture and serves as a diagnostic tool to verify logical consistency.

6.1 Gradient Field Heatmap

The first output is a heatmap representing the Biological Gradient Field. This heatmap shows the spatial distribution of signal intensity across the two dimensional environment. The highest intensity is clearly concentrated at the centre of the domain, representing the tumour like region, while intensity decreases smoothly as distance from the centre increases.

This output confirms that the environment layer is correctly implemented. The smooth radial decay verifies that the gradient field function behaves as expected and provides a consistent signal landscape for nanoswimmer sensing.

6.2 Nanoswimmers Position Scatter Plot



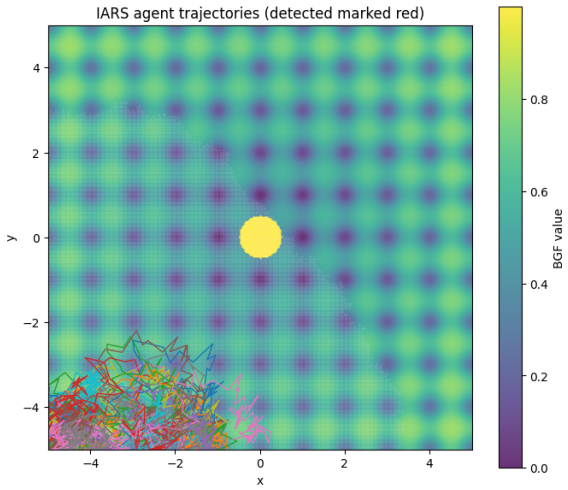
Overlaid on the gradient field, the simulation displays a scatter plot of nanoswimmer positions. Each point represents the location of an individual nanoswimmer at the sensing phase.

The random spatial distribution confirms unbiased initialization and ensures that nanoswimmers experience diverse local gradient conditions. This output validates that agents are not pre clustered or artificially biased toward the target region.

6.3 Local Gradient Direction Arrows

From each nanoswimmer position, arrows are drawn to represent locally estimated gradient directions. These arrows point in the direction of increasing signal intensity as computed through finite difference sampling. Inspection of these arrows shows that, despite some variability and noise, most directions generally orient toward the central high intensity region. This confirms that the finite difference gradient approximation is functioning correctly and that nanoswimmers are able to infer meaningful directional cues using only local information.

6.4 Elite Agent Highlighting



Elite nanoswimmers selected based on fitness ranking are visually distinguished from non-elite agents. These highlighted agents are typically located closer to the central region where gradient intensity is stronger. This output confirms that the fitness evaluation and elite selection logic is correctly filtering agents based on sensing reliability. The spatial concentration of elite agents near stronger signal regions aligns with theoretical expectations of the IARS framework.

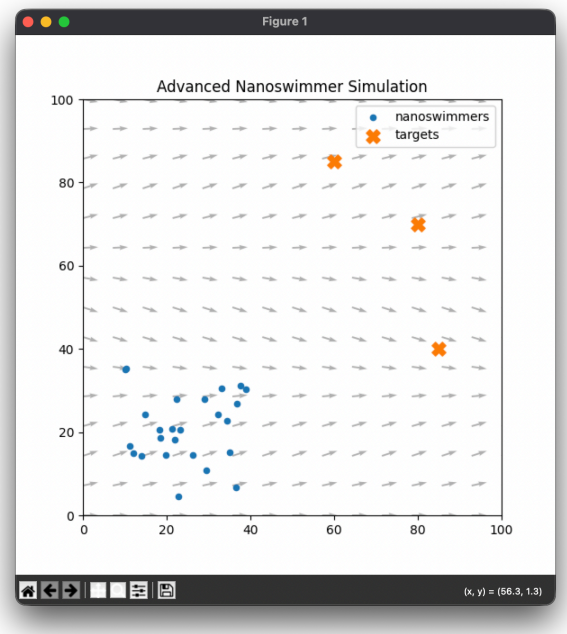
6.5 Fused Global Direction Arrow

A single prominent arrow is drawn to represent the final fused global navigation direction. This arrow is computed using weighted vector fusion of elite agent directions. The orientation of this arrow consistently points toward the tumour like centre of the gradient field. This confirms that the direction fusion module is correctly aggregating elite information and suppressing noise from unreliable agents. The smoothness and stability of this arrow demonstrate the effectiveness of vector based fusion over naive angular averaging.

6.6 Numerical Output of Final Direction

In addition to visual arrows, the simulation outputs numerical values corresponding to the final fused direction angle. These values provide quantitative confirmation that the global direction aligns with the expected gradient ascent direction. The consistency between numerical direction values and visual arrow orientation further validates the correctness of the implementation.

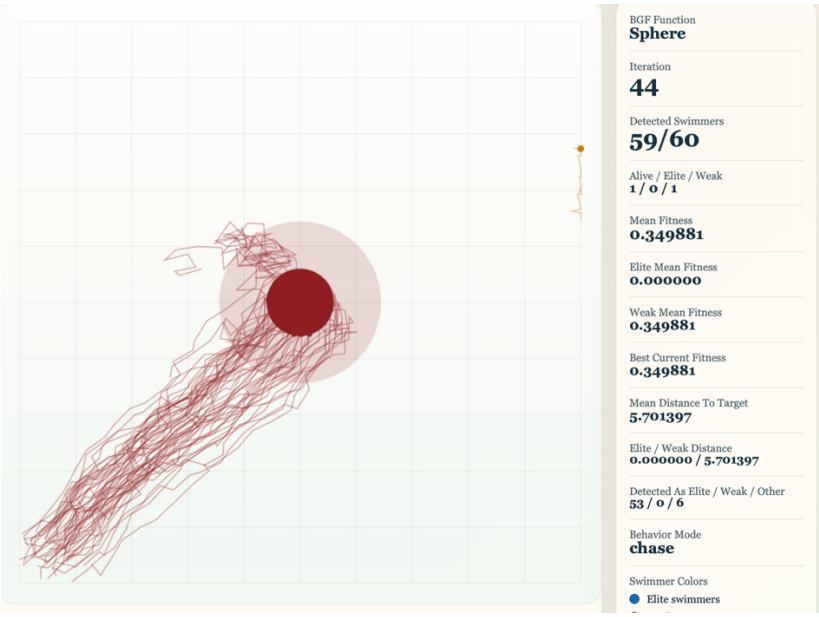
Overall Validation



Taken together, the heatmaps, scatter plots, directional arrows, elite highlighting, and numerical outputs form a comprehensive validation suite. They collectively demonstrate that the simulation correctly implements biological gradient modelling, nanoswimmer sensing, fitness based filtering, and collective decision formation. These outputs confirm that the first computational half of the Individual Adaptive Regulation Strategy is implemented accurately and functions as intended. The validated framework is therefore well prepared for future extensions involving motion execution, adaptive behaviours, clustering, and performance analysis under more complex biological conditions.

7. Testing and Validation

Validation of the implemented IARS inspired simulation was carried out using a combination of visual inspection, numerical consistency checks, and behavioural verification. This multi-level validation approach ensures that both individual algorithmic components and overall system behaviour operate as intended.



Visual inspection formed the first level of validation. Gradient heatmaps consistently displayed a smooth and symmetric intensity distribution with a clear maximum at the centre of the environment, representing the tumour like region. Directional arrows originating from nanoswimmer positions were observed to generally orient toward this central region, confirming that finite difference based gradient sensing correctly captures local directional information. The final fused direction arrow was consistently aligned with the global gradient ascent direction, providing immediate visual confirmation of correct collective decision making. Numerical consistency checks provided quantitative validation. Fitness values computed for nanoswimmers increased as their positions approached the centre of the gradient field, confirming that fitness accurately reflects signal strength. The indices of elite nanoswimmers selected through fitness ranking consistently corresponded to agents located in higher intensity regions. Additionally, the numerical value of the fused global direction angle matched the orientation observed in the visual plots, confirming consistency between computational output and graphical representation. Behavioural verification served as the final and most important validation layer. Across multiple simulation runs with different random initializations, elite nanoswimmers consistently emerged from regions with stronger biological signals. Despite variability in individual gradient estimates, the weighted vector fusion process produced a stable

and reliable global direction in every run. This stability demonstrates the effectiveness of the IARS principle of selective information aggregation and confirms robustness against local sensing noise.

Together, these validation results confirm that the biological gradient modelling, nanoswimmer sensing, fitness evaluation, elite selection, and direction fusion modules are all functioning correctly. The consistency observed across visual, numerical, and behavioural checks establishes confidence in the correctness of the implemented framework and verifies that the early computational stages of the Individual Adaptive Regulation Strategy are accurately realised.

8. Conclusion

This project successfully implements and validates the foundational sensing and decision making stages of an Individual Adaptive Regulation Strategy inspired nanoswimmer framework within a controlled two dimensional simulation environment. By isolating and rigorously testing gradient sensing, fitness evaluation, elite selection, and weighted directional fusion, the work ensures that the core computational logic of the strategy is both correct and reliable.

The results demonstrate that collective sensing, when combined with selective trust in high quality agents, produces stable and accurate navigation directions even in the presence of local estimation variability. Elite selection consistently prioritises nanoswimmers located in stronger signal regions, while weighted vector fusion effectively suppresses noise and transforms individual measurements into coherent swarm level intelligence. Most importantly, the observed stability of the fused direction across multiple runs confirms the robustness of the IARS decision making principle. This validates the strategy's suitability for biologically realistic environments where uncertainty and noise are unavoidable.

Overall, the project establishes a strong and well verified computational foundation for future extensions. With the sensing and decision making pipeline confirmed, subsequent work can confidently incorporate motion execution, adaptive behavioural switching, clustering based coordination, and performance evaluation under dynamic conditions. This framework therefore represents a meaningful step toward intelligent nanoscale navigation systems for biomedical applications.

9. Future Scope

• Implement full sensing and driving cycles where nanoswimmers alternate between environmental perception and motion execution based on the fused navigation direction

• Introduce clustering based chase behaviour using DBSCAN to enable local group coordination and cooperative navigation toward high confidence signal regions

• Develop probabilistic behavioural switching mechanisms allowing nanoswimmers to transition between exploration, exploitation, and regrouping states based on sensing confidence and swarm consensus

• Integrate magnetic field driven motion models to study interaction between external actuation and internally computed swarm intelligence

• Define robust termination conditions such as proximity to tumour centre, convergence stability thresholds, or energy based stopping criteria

• Establish quantitative performance metrics including convergence time, navigation accuracy, swarm dispersion, robustness to noise, and efficiency

• Add interactive dashboards and real time charts to visualise swarm convergence, decision stability, and temporal behaviour during simulation runs.

10. References

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